

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
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Game Theory and Strategy

Section Id :	640653141362
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	6
Number of Questions to be attempted :	6
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653398355
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532166313 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : GAME THEORY AND STRATEGY

(COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406537043761. ✓ Yes

6406537043762. ✗ No

Sub-Section Number :

2

Sub-Section Id :

640653398356

Question Shuffling Allowed :

No

Question Id : 6406532166314 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (2 to 3)

Question Label : Comprehension

Consider a duopoly game in which two firms simultaneously and independently select prices that are greater than or equal to zero. Denote firm 1's price as P_1 and firm 2's price as P_2 . After the prices are set, consumers demand $10 - P_1 + P_2$ units of the good that firm 1 produces, and they demand $10 - P_2 + P_1$ units of the good that firm 2 produces. Assume that each firm produces at zero cost, so firm i 's payoff (profit) is $(10 - P_i + P_j)P_i$, where P_i is firm i 's price and P_j is the other firm's price. Answer the given sub questions based on this information:

Sub questions

Question Number : 2 Question Id : 6406532166315 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

$BR_i(P_j)$ will be

Options :

6406537043763. ✗ $5 + \frac{P_j}{3}$

6406537043764. ✗ $5 + \frac{2P_j}{3}$

6406537043765. ✓ $5 + \frac{P_j}{2}$

6406537043766. ✖ $5 + P_j$

Question Number : 3 Question Id : 6406532166316 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Choose the incorrect statement

Options :

6406537043767. ✖ This game is a symmetric game

6406537043768. ✖ This game has a symmetric Nash equilibrium

6406537043769. ✔ Under the Nash equilibrium each player chooses $P=15$

6406537043770. ✖ This game has unique Nash equilibrium

Question Id : 6406532166329 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (4 to 5)

Question Label : Comprehension

An item is up for auction. Player 1 values the item at 5 while player 2 values the item at 4. Each player can bid either 0, 1 or 2. If player i bids more than player j then i wins the item and pays his bid, while the loser does not pay. If both players bid the same amount, then a coin is tossed to determine who the winner is, who gets the item and pays his bid while the loser pays nothing. Formulate the game in matrix form and answer the given sub questions.

Sub questions

Question Number : 4 Question Id : 6406532166330 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which player does not have strictly dominated strategy?

Options :

6406537043808. ✖ Player 1

6406537043809. ✔ Player 2

6406537043810. ✖ None

Question Number : 5 Question Id : 6406532166331 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which strategies survive Iterated Elimination of Strictly Dominated Strategy?

Options :

6406537043811. ✖ (0, 0)

6406537043812. ✖ (1, 1)

6406537043813. ✔ (2, 2)

6406537043814. ✖ None

Question Id : 6406532166332 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (6 to 7)

Question Label : Comprehension

(Synergies) Two division managers can invest time and effort in creating a better working relationship. Each invests $e_i \geq 0$, and if both invest more then both are better off, but it is costly for each manager to invest. In particular, the payoff function for player i from effort levels (e_i, e_j) is $v_i(e_i, e_j) = (a + e_j)e_i - 2e_i^2$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 6 Question Id : 6406532166333 Question Type : MCQ Correct Marks : 2

Question Label : Multiple Choice Question

What is the best response function of player i for all $e_i \geq 0$

Options :

6406537043815. ✔ $BR_i(e_j) = (a + e_j)/4$

6406537043816. ✖ $BR_i(e_i) = (a + e_i)/2$

6406537043817. ✖ $BR_i(e_j) = (a + 2e_i)/4$

6406537043818. ✖ $BR_i(e_i) = (a + 2e_j)/4$

Question Number : 7 Question Id : 6406532166334 Question Type : MCQ Correct Marks : 2

Question Label : Multiple Choice Question

Choose the correct alternative

Options :

6406537043819. ✖ Under Nash equilibrium both players choose $e_1 = e_2 = a/3$.

6406537043820. ✖ This is a unique Nash equilibrium

6406537043821. ✔ Both Under Nash equilibrium both players choose $e_1 = e_2 = a/3$ & This is a unique Nash equilibrium

6406537043822. ✖ None

Sub-Section Number : 3
Sub-Section Id : 640653398357
Question Shuffling Allowed : No

Question Id : 6406532166317 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (8 to 9)

Question Label : Comprehension

Consider the following normal form game and answer the given sub questions:

	Player 2		
		Left	Right
	Top	5,4	0,3
	Bottom	3,2	1,7

Sub questions

Question Number : 8 Question Id : 6406532166318 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

This game has

Options :

6406537043771. ✔ Two Nash equilibria in pure strategy

6406537043772. ✖ A unique Nash equilibrium in pure strategy

6406537043773. ✖ Both Two Nash equilibria in pure strategy & A unique Nash equilibrium in pure strategy are true

6406537043774. ✖ None is true

Question Number : 9 Question Id : 6406532166319 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 3

Question Label : Short Answer Question

If under mixed strategy equilibrium, let p and q denote the probability of player 1's choosing 'Top' and player 2's choosing 'Left', respectively, value of $\frac{1}{p-q}$ will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Sub-Section Number : 4

Sub-Section Id : 640653398358

Question Shuffling Allowed : No

Question Id : 6406532166320 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (10 to 17)

Question Label : Comprehension

Consider a two-player game in which player 1 can choose A or B. The game ends if he chooses A while it continues to player 2 if he chooses B. Player 2 can either choose C or D, with the game ending after C and continuing again with player 1 after D. Player 1 then can choose E or F, and then the game ends after each of these choices.

Imagine that the payoffs following choice A by player 1 are (2, 0), following C by player 2 are (3, 2), following E by player 1 are (2, 1) and following F by player 1 are (1, 2)

Model this as an extensive form game and answer the given problems:

Sub questions

Question Number : 10 Question Id : 6406532166321 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

This is a game of

Options :

6406537043776. ✓ Perfect information

6406537043777. ✗ Imperfect information

6406537043778. ✗ None

6406537043779. ✖ Can't be determined

Question Number : 11 Question Id : 6406532166322 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many terminal nodes does this game have?

Options :

6406537043780. ✖ 3

6406537043781. ✖ 2

6406537043782. ✔ 4

6406537043783. ✖ 5

Question Number : 12 Question Id : 6406532166323 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many information sets does this game have?

Options :

6406537043784. ✖ 2

6406537043785. ✔ 3

6406537043786. ✖ 4

6406537043787. ✖ 1

Question Number : 13 Question Id : 6406532166324 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many pure strategies does player 1 have?

Options :

6406537043788. ✖ 1

6406537043789. ✖ 2

6406537043790. ✖ 3

6406537043791. ✔ 4

Question Number : 14 Question Id : 6406532166325 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many pure strategies does player 2 have?

Options :

6406537043792. ✖ 1

6406537043793. ✔ 2

6406537043794. ✖ 3

6406537043795. ✖ 4

Question Number : 15 Question Id : 6406532166326 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many pure strategy Nash equilibria does this game have?

Options :

6406537043796. ✖ 1

6406537043797. ✖ 2

6406537043798. ✖ 3

6406537043799. ✔ 4

Question Number : 16 Question Id : 6406532166327 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which one of the Nash equilibria is also a subgame perfect equilibrium?

Options :

6406537043800. ✖ (AE, D)

6406537043801. ✖ (AF, D)

6406537043802. ✔ (BE, C)

6406537043803. ✖ (BF, C)

Question Number : 17 Question Id : 6406532166328 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

No. of sub-games this game has

Options :

6406537043804. ✖ 1

6406537043805. ✖ 2

6406537043806. ✔ 3

6406537043807. ✖ 4